

Total No. of Questions : 8]

SEAT No. :

PC2778

[Total No. of Pages : 3

[6352]-2

S.E. (Civil)

SURVEY

(2019 Pattern) (Semester - IV) (201009)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4 or Q.5 or Q.6, Q.7 & Q.8.
- 2) Neat sketches must be drawn wherever necessary.
- 3) Figures to right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of electronic pocket calculator is allowed.
- 6) Use of cell phone is prohibited in examination hall.

Q1) a) Explain principle of stadia tacheometry and methods for determination of constants. [5]

b) A Tacheometer was set up at Station at C and the following readings were obtained on a staff vertically held. Determine RL of point D and distance CD when Constants are 100 & 0.15. RL of BM = 750.50 m [7]

Inst Station	Staff Station	Vertical Angle	Hair Reading
C	BM	-5°20'	1.500,1.800,2.450
C	D	+8°12'	0.750,1.500,2.250

c) State characteristics of contours. [6]
OR

Q2) a) Define Contour Interval, Horizontal Equivalent and state uses of contour maps. [5]

b) The following observations were made using a tacheometer fitted with an anallatic lens, Staff held vertically and multiplying constant being 100. Determine RL of point B and distance Between A & B. RL of BM = 255.75 m [7]

Inst Station	HI	Staff Station	Vertical Angle	Hair Reading
P	1.255	BM	-4°20'	1.325,1.825,2.325
P	1.255	A	+6°30'	0.850,1.600,2.350
B	1.450	A	-7°24'	1.715,2.315,2.915

c) Enlist different indirect methods of contouring. Explain any one method with detailed sketch. [6]

P.T.O.

- Q3)** a) Draw a neat sketch of Simple Circular curve and Properties of Simple Circular Curves. [5]
- b) Two straight roads intersect at a chainage of 150.5m. The angle of deflection being 30° . Taking radius of 100 m, calculate necessary data for setting curve by Methods of Offsets from Long Chords. [7]
- c) Write a note on necessity and types of transition curves. [5]

OR

- Q4)** a) What is transition curve, state the applications of transition curve? [5]
- b) Two straights AB and BC meet at chainage of 1250 m. A right-handed simple circular curve of 250 m radius joins them. The Intersection angle between two straights is 150° . Tabulate the necessary data to layout the curve by Rankine's method of deflection angle. Take Peg Interval as 20 m and Least Count 20". [7]
- c) Explain Horizontal Curves and Types of Horizontal Curve. [5]
- Q5)** a) Explain Segments and advantages of Space Based Positioning System. [6]
- b) Explain necessity of horizontal and vertical controls in construction activity. [5]
- c) Write a note on Construction Survey. [6]

OR

- Q6)** a) State Different names of satellites and Explain any One in details. [6]
- b) Write a short note on survey for drainage line work. [6]
- c) Write a short note on Shore Line Survey. [5]

- Q7)** a) What do you mean by triangulation and trilateration in geodetic survey?[6]
b) Define Sounding and state any one method of sounding with sketch.[6]
c) Differentiate between Map and aerial photograph. [6]

OR

- Q8)** a) State the working principle and applications of total station. [6]
b) Differentiate between Plane and Geodetic Survey. [6]
c) Explain sounding methods and sounding equipment of hydrographic survey. [6]

